

# Replay-Aware TSP Benchmark Report

## Overview

- Condition count: 4.
- Best final transfer gap: tsplib\_random\_replay.
- Best TSPLIB holdout gap: tsplib\_random\_replay.
- Best synthetic holdout gap: tsplib\_random\_replay.

## Run Metadata

- run\_name: run\_20260513\_135156\_h
- started\_at\_local: 2026-05-13 13:51:56
- finished\_at\_local: 2026-05-13 14:01:45
- duration\_hhmm: 00:10
- duration\_seconds: 589.216
- seed\_offset: 7000
- replicate\_label: h
- judge\_status: enabled
- judge\_provider: openai
- judge\_model: gpt-4.1-mini

## Condition Comparison

| Condition                         | Replay  | Selection    | Compression | Final TSPLIB Gap | Final Synthetic Gap | Final Transfer Gap | Mean Accepted Novelty | Mean Complexity | Adaptation Efficiency |
|-----------------------------------|---------|--------------|-------------|------------------|---------------------|--------------------|-----------------------|-----------------|-----------------------|
| tsplib_no_replay                  | none    | score_only   | False       | 0.213387         | 0.064916            | 0.159398           | 0.882029              | 0.56            | -0.028236             |
| tsplib_random_replay              | random  | score_only   | False       | 0.073344         | 0.006169            | 0.048917           | 0.672212              | 0.76            | 0.052384              |
| tsplib_failure_replay             | failure | score_only   | False       | 0.213387         | 0.064916            | 0.159398           | 0.78455               | 0.56            | -0.031744             |
| tsplib_failure_replay_compression | failure | novelty_gate | True        | 0.173003         | 0.017074            | 0.116302           | 0.817808              | 0.76            | 0.06617               |

## Condition Notes

### tsplib\_no\_replay

- Replay mode: none with selection mode score\_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 3, mean accepted code novelty 0.882029, and final complexity 0.56.
- Adaptation efficiency -0.028236 and archive sizes {'worst\_cases': 6, 'failure\_cases': 6, 'adversarial\_layouts': 4} / elite 1.
- Panel heldout\_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic\_holdout mean gap 0.064916 across 4 instances; family means: clustered\_gaussian=0.0, grid\_outliers=0.259662, ring\_bridge=0.0, two\_corridors=0.0.
- Worst recent training cases: [{"best\_known\_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality\_gap": 0.401318}, {"best\_known\_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality\_gap": 0.391097}, {"best\_known\_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality\_gap": 0.328813}]

### **tsplib\_random\_replay**

- Replay mode: random with selection mode score\_only.
- Final transfer gaps: TSPLIB 0.073344, synthetic 0.006169, combined 0.048917.
- Accepted-epoch count 3, mean accepted code novelty 0.672212, and final complexity 0.76.
- Adaptation efficiency 0.052384 and archive sizes {'worst\_cases': 6, 'failure\_cases': 6, 'adversarial\_layouts': 4} / elite 3.
- Panel heldout\_tsplib mean gap 0.073344 across 7 instances; family means: ch=0.039647, kroD=0.074199, pcb=0.176592, pr=0.043621, rd=0.081922, st=0.057778.
- Panel synthetic\_holdout mean gap 0.006169 across 4 instances; family means: clustered\_gaussian=0.024674, grid\_outliers=0.0, ring\_bridge=0.0, two\_corridors=0.0.
- Worst recent training cases: [{"best\_known\_cost": 7542, "cost": 10061, "family": "berlin", "name": "berlin52", "optimality\_gap": 0.333996}, {"best\_known\_cost": 14379, "cost": 19018, "family": "lin", "name": "lin105", "optimality\_gap": 0.322623}, {"best\_known\_cost": 2579, "cost": 3404, "family": "a", "name": "a280", "optimality\_gap": 0.319891}]

### **tsplib\_failure\_replay**

- Replay mode: failure with selection mode score\_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.78455, and final complexity 0.56.
- Adaptation efficiency -0.031744 and archive sizes {'worst\_cases': 6, 'failure\_cases': 6, 'adversarial\_layouts': 4} / elite 1.
- Panel heldout\_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic\_holdout mean gap 0.064916 across 4 instances; family means: clustered\_gaussian=0.0, grid\_outliers=0.259662, ring\_bridge=0.0, two\_corridors=0.0.
- Worst recent training cases: [{"best\_known\_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality\_gap": 0.401318}, {"best\_known\_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality\_gap": 0.391097}, {"best\_known\_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality\_gap": 0.328813}]

### **tsplib\_failure\_replay\_compression**

- Replay mode: failure with selection mode novelty\_gate.
- Final transfer gaps: TSPLIB 0.173003, synthetic 0.017074, combined 0.116302.

- Accepted-epoch count 2, mean accepted code novelty 0.817808, and final complexity 0.76.
- Adaptation efficiency 0.06617 and archive sizes {'worst\_cases': 6, 'failure\_cases': 6, 'adversarial\_layouts': 4} / elite 1.
- Panel heldout\_tsplib mean gap 0.173003 across 7 instances; family means: ch=0.16062, kroD=0.169203, pcb=0.23262, pr=0.199937, rd=0.19469, st=0.093333.
- Panel synthetic\_holdout mean gap 0.017074 across 4 instances; family means: clustered\_gaussian=0.006511, grid\_outliers=0.061786, ring\_bridge=0.0, two\_corridors=0.0.
- Worst recent training cases: [{"best\_known\_cost": 2579, "cost": 3451, "family": "a", "name": "a280", "optimality\_gap": 0.338116}, {"best\_known\_cost": 7542, "cost": 9545, "family": "berlin", "name": "berlin52", "optimality\_gap": 0.265579}, {"best\_known\_cost": 14379, "cost": 17825, "family": "lin", "name": "lin105", "optimality\_gap": 0.239655}]

## Judge Appendix

### Summary of Replay-Aware TSP Benchmark Results

| Condition                          | Replay Mode           | Selecti on Mode | Mean TSPLIB Gap   | Mean Synthetic Gap | Mean Transfer Gap | Code Novelty (Mean) | Com plexi ty | Adaptation Efficiency |
|------------------------------------|-----------------------|-----------------|-------------------|--------------------|-------------------|---------------------|--------------|-----------------------|
| tsplib_no_replay                   | none                  | score_o nly     | 0.2134            | 0.0649             | 0.1594            | 0.8820              | 0.56         | -0.0282               |
| tsplib_random_replay               | random                | score_o nly     | <b>**0.0733**</b> | <b>**0.0062**</b>  | <b>**0.0489**</b> | 0.6722              | 0.76         | 0.0524                |
| tsplib_failure_replay              | failure               | score_o nly     | 0.2134            | 0.0649             | 0.1594            | 0.7846              | 0.56         | -0.0317               |
| tsplib_failure_replay_c ompression | failure + compression | novelty _gate   | 0.1730            | 0.0171             | 0.1163            | 0.8178              | 0.76         | 0.0662                |

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### Interpretation

- **\*\*Best transfer performance\*\*** (lowest TSPLIB and synthetic holdout gaps) is given by **\*\*tsplib\_random\_replay\*\***:
- Mean TSPLIB gap (0.0733) and mean synthetic gap (0.0062) substantially outperform all other conditions.
- Mean transfer gap (0.0489) is also the lowest, indicating strongest generalization to held-out and synthetic data.
- **\*\*Code novelty\*\*** decreases in tsplib\_random\_replay (mean 0.6722) compared to no replay (0.8820), despite improved transfer metrics.
- This suggests that improved transfer is achieved **\*\*not by increasing code novelty\*\***, but by more effective reuse via random replay.
- **\*\*No replay\*\*** and **\*\*failure replay\*\*** without compression have identical final gaps, showing no transfer improvement and negative adaptation efficiencies.
- The **\*\*failure replay with compression and novelty gate\*\*** has some transfer improvement over no replay/failure replay:
  - It reduces mean TSPLIB gap from 0.2134 to 0.1730 and synthetic gap from 0.0649 to 0.0171.
  - Adaptation efficiency is positive (0.0662), but transfer remains worse than random replay.

- Code novelty (0.8178) is intermediate but still higher than random replay.
- Compression pressure is active here, indicating an added constraint during replay.
- **Complexity** is higher (~0.76) in random replay and failure replay with compression compared to 0.56 in no replay and failure replay without compression.

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## Conservative Conclusions

- **Random replay** yields the strongest evidence of improved generalization and transfer based on both the held-out TSPLIB and synthetic holdout gaps.
- The improvement in transfer performance with random replay occurs alongside a **drop in code novelty**, meaning better performance is due to effective exploitation rather than new algorithmic invention.
- Failure replay without compression does not improve transfer relative to no replay, indicating that replay type matters.
- Adding compression pressure with failure replay improves transfer gaps somewhat but fails to outperform random replay.
- Complexity increases moderately with replay modes that facilitate transfer, reflecting potentially richer behavior profiles.
- Adaptation efficiency correlates positively with improved transfer: random replay and failure replay with compression have positive adaptation efficiencies, while others are negative.

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## Key Takeaways

- Prioritize **random replay** for maximizing transfer and optimality-gap improvements in this replay-aware TSP benchmark.
- Code novelty metrics alone are insufficient to claim algorithmic innovation; lower novelty with better transfer suggests effective learning or memory utilization instead.
- Different replay modes distinctly impact transfer, underlining the need to differentiate no replay, random replay, failure replay, and compression-aware replay in analysis.
- Compression-aware failure replay partially closes the transfer gap but does not surpass random replay, indicating room for further methods combining replay types and complexity control.